ON A RINGED SITE THE O-MODULES THAT ARE SUMMANDS OF FINITE FREE O-MODULES FORM A CATEGORY EQUIVALENT TO THE FINITE PROJECTIVE R-MODULES FOR R THE RING OF GLOBAL SECTIONS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Same-name restatement check: the label lemma-category-summands-finite-free also occurs in the chapter cohomology.tex (ringed spaces, a special case of the ringed-site setting) and is there a special case, so the more general ringed-site statement is the one counted for this lane.

Editorial note. The label lemma-category-summands-finite-free defines the statement; the counted proof is the proof environment immediately following the lemma in the source.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Global sections functor on a ringed site. Same chapter, the section functor Gamma(C, -) on O-modules of a ringed site (C, O) with values in R-modules, R = Gamma(C, O). Supplies the functor E -> Gamma(C, E) and its essential surjectivity target.
- Constant sheaf of a set on a point topos and the morphism of ringed topoi. Same chapter, the pushforward f_* = global sections and the constant-sheaf pullback f^{-1} defining the morphism of ringed topoi (Sh(C), O) -> (Sh(pt), R). Supplies the inverse functor f^* = Gamma(C, -) tensored with O, used to reconstruct a module from its finite projective sections.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Set $R = \Gamma(\mathcal{C}, \mathcal{O})$. The category of $\mathcal{O}$-modules which are summands of finite free $\mathcal{O}$-modules is equivalent to the category of finite projective R-modules.*

Proof. Observe that a finite projective R-module is the same thing as a summand of a finite free R-module. The equivalence is given by the functor $\mathcal{E} \mapsto \Gamma(\mathcal{C}, \mathcal{E})$. The inverse functor is given by the following construction. Consider the morphism of topoi $f : Sh(\mathcal{C}) \to Sh(\text{pt})$ with f_* given by taking global sections and f^{-1} by sending a set S, i.e., an object of $Sh(\text{pt})$, to the constant sheaf with value S. We obtain a morphism $(f, f^\sharp) : (Sh(\mathcal{C}), \mathcal{O}) \to (Sh(\text{pt}), R)$ of ringed topoi by using the identity map $R \to f_*\mathcal{O}$. Then the inverse functor is given by f^*. □

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